ESMValTool

Earth System Model Evaluation Tool

ESMValTool Advanced Topics I

Online Tutorial

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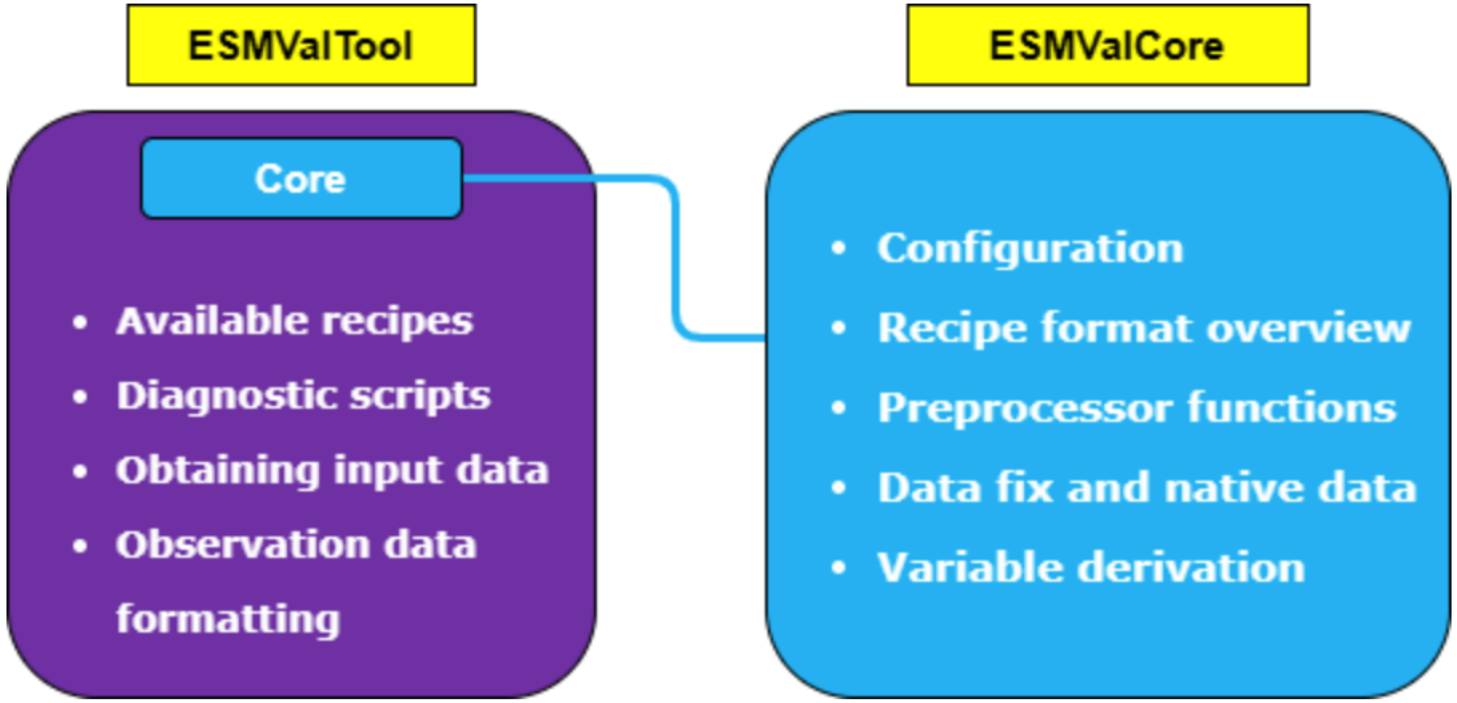
Outline

- Introductions – talk will be recorded.
- Recap – Basic components, recipes and development mode.
- Diagnostics – theory
- ESMValTool for the community
- How to get help
- Questions
- Break
- Hands on session

If you are completely new to ESMValTool

- Please refer to the tutorial and slides from the basic topics tutorial to catch up
- Tutorial : https://tutorial.esmvaltool.org/index.html
- Slides from previous tutorials: https://github.com/ESMValGroup/Community#training-events

ESMValTool – Components



ESMValTool Recipes

ESMValTool is controlled through a **recipe** (text file in YAML format).

Recipes contain

- Datasets to analyze
- Preprocessors to use
- Variables to analyze
- Diagnostic scripts to use (Python, NCL or R scripts)

ESMValTool Configuration – New!

- Configuration files are YAML files.
- They specify configuration options for running ESMValTool – where to put output, where to download data etc.
- Changes made since v2.12 to support nested configuration structure.
- Key for today's exercises (essential to do):
  - **~/.config/esmvaltool – default location for config files**.
  - Basic set of config files needed for today – config-user.yml, data-hpc-badc.yml or data-hpc-dkrz.yml (assuming use of JASMIN/Levante).
  - Optionally data-intake-esgf.yml and intake-esgf/conf.yml
  - Tutorial episode available - please follow for guidance.

ESMValTool Configuration (config-user.yml)

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```yaml
---
# Destination directory where all output will be written
# Includes log files and performance stats.
output_dir: /work/scratch-nopw2/user/esmvaltool_output/


# Log level of the console --- debug/[info]/warning/error
# For much more information printed to screen set log_level to ``debug``.
log_level: info


# Plot file format --- [png]/pdf/ps/eps/epsi
output_file_type: png

# Remove the ``preproc`` directory if the run was successful --- [true]/false
# By default this option is set to ``true``, so all preprocessor output files
# will be removed after a successful run. Set to ``false`` if you need those
# files.
remove_preproc_dir: false
```

Change output directory to something not in your home directory – avoids exceeding quotas.

Useful to have more informative and detailed log output when starting or debugging!

Available plot file formats and current default.

Retain preprocessed output – useful when starting!

ESMValTool Configuration

Small change for today's tutorial. In data-hpc-badc.yml under CMIP5: Change:

```
dirname_template: "{project.lower}/{product}/{institute}/{dataset}/{exp}/{frequency}/{modeling_realm}/{mip}/{ensemble}/{version}"
```

to

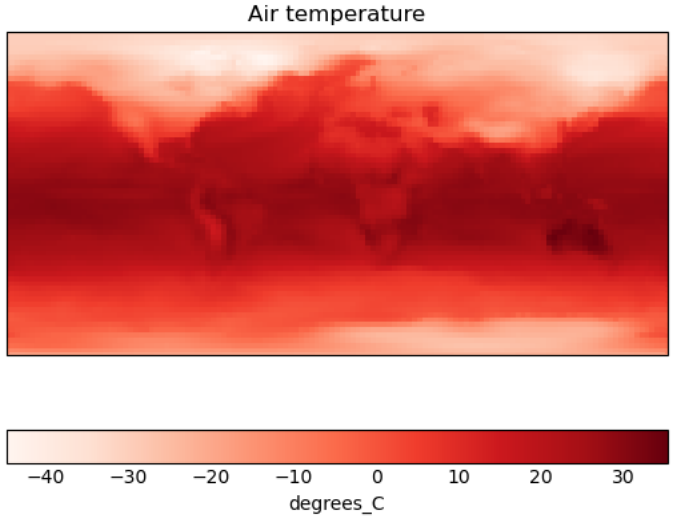
```
dirname_template: "{project.lower}/{product}/{institute}/{dataset}/{exp}/{frequency}/{modeling_realm}/{mip}/{ensemble}/{version}/{short_name}"
```

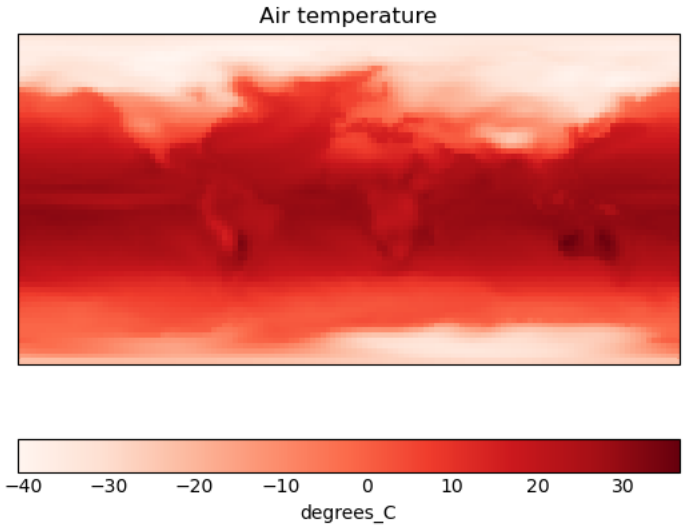
Elements of the recipe

```yaml
datasets:
  - {dataset: BCC-ESM1, project: CMIP6, exp: historical, ensemble: r1i1p1f1, grid: gn}
  - {dataset: bcc-csm1-1, version: v1, project: CMIP5, exp: historical, ensemble: r1i1p1}

diagnostics:

  map:
    description: Global map of temperature in January 2000.
    themes:
      - phys
    realms:
      - atmos
    variables:
      tas:
        mip: Amon
        preprocessor: to_degrees_c
        timerange: 2000/P1M
        caption: |
          Global map of {long_name} in January 2000 according to {dataset}.
```

```yaml
to_degrees_c:
  convert_units:
    units: degrees_C
```

Diagnostic Output





Running ESMValTool

- Before running ESMValTool:
  - load module or activate conda environment.
  - Ensure config files are in place
  - Run example recipe as per Tutorial lesson on writing own diagnostic.

- If *run* is successful, in the esmvaltool_output directory, files and subdirectories will be created under recipe_python_#_#:
  - index.html
  - run
  - preproc
  - plots
  - Work

- This directory structure has files (preprocessor output, data specifications etc.) for diagnostic Python script to read.

Where does the output go?

For preprocessed output:

output directory -> preproc -> map (diagnostic name)

->tas (variable name)

which has:

- processed files (netCDF)
- provenance files – information about the data and how it was generated (xml files)
- **metadata.yml** file – an entry for every variable in every diagnostic – interface file for preprocessor output.

Information about data

```yaml
datasets:
  - {dataset: BCC-ESM1, project: CMIP6, exp: historical, ensemble: r1i1p1f1, grid: gn}
  - {dataset: bcc-csm1-1, version: v1, project: CMIP5, exp: historical, ensemble: r1i1p1}
```

```yaml
diagnostics:

  map:
    description: Global map
    themes:
      - phys
    realms:
      - atmos
    variables:
      tas:
        mip: Amon
        preprocessor: to_degrees_c
        timerange: 2000/PIM
        caption: |
          Global map of {long_name} in January 2000 according to {dataset}.
```

```yaml
        Global map of {long_name} in January 2000 according to {dataset}.
    scripts:
      script1:
        script: /path/ESMValTool/esmvaltool/diag_scripts/examples/diagnostic.py
        quickplot:
          plot_type: pcolormesh
          cmap: Reds
```

```yaml
to_degrees_c:
  convert_units:
    units: degrees_C
```

Preprocessor Interface file

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```yaml
? /outputdir/recipe_python_#_#/preproc/map/tas/CMIP6_BCC-ESM1_Amon_historical_r1i1p1f1_tas_gn_20000101-20000201.nc
: activity: CMIP
  alias: CMIP6
  caption: 'Global map of {long_name} in January 2000 according to {dataset}.

    '
  dataset: BCC-ESM1
  diagnostic: map
  end_year: 2000
  ensemble: r1i1p1f1
  exp: historical
  filename: /outputdir/recipe_python_#_#/preproc/map/tas/CMIP6_BCC-ESM1_Amon_historical_r1i1p1f1_tas_gn_20000101-20000201.nc
  frequency: mon
  grid: gn
  institute:
  - BCC
  long_name: Near-Surface Air Temperature
  mip: Amon
  modeling_realm:
  - atmos
  preprocessor: to_degrees_c
  project: CMIP6
  recipe_dataset_index: 0
  short_name: tas
  standard_name: air_temperature
  start_year: 2000
  timerange: 2000/P1M
  units: degrees_C
  variable_group: tas
  version: v20181214
```

Where does the output go?

For diagnostic output:

output directory -> work/plots/ -> map (diagnostic name)

->script1 (script name in recipe file)

which have:

- processed files (netCDF) or plots
- provenance files – information about output and how it was generated (xml files)
- citation information

Where does the output go?

For diagnostic output:

output directory -> run ->

which has:

- log files, resource usage information
- map (diagnostic)
    - provenance file
    - diagnostic specific log
    - resource usage
    - **settings.yml** - interface file

Diagnostic Interface file

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```yaml
quickplot:
  plot_type: pcolormesh
  cmap: Reds
recipe: recipe_python.yml
version: 2.14.0
script: script1
run_dir: /outputdir/recipe_python_#_#/run/map/script1
plot_dir: /outputdir/recipe_python_#_#/plots/map/script1
work_dir: /outputdir/recipe_python_#_#/work/map/script1
output_file_type: png
log_level: info
auxiliary_data_dir: /auxdatapath/auxiliary_data
themes:
- phys
realms:
- atmos
input_files:
- /outputdir/recipe_python_#_#/preproc/map/tas/metadata.yml
```

Helpful utility functions

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```python

from esmvaltool.diag_scripts.shared import (
    group_metadata,
    run_diagnostic,
    save_data,
    save_figure,
    select_metadata,
    sorted_metadata,
)
from esmvaltool.diag_scripts.shared.plot import quickplot

```

Helpful utility functions

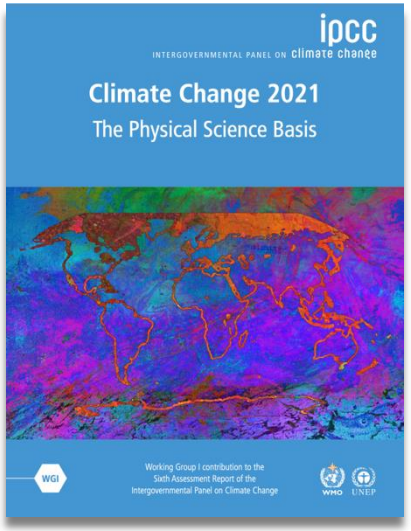
ESMValTool
Earth System Model Evaluation Tool

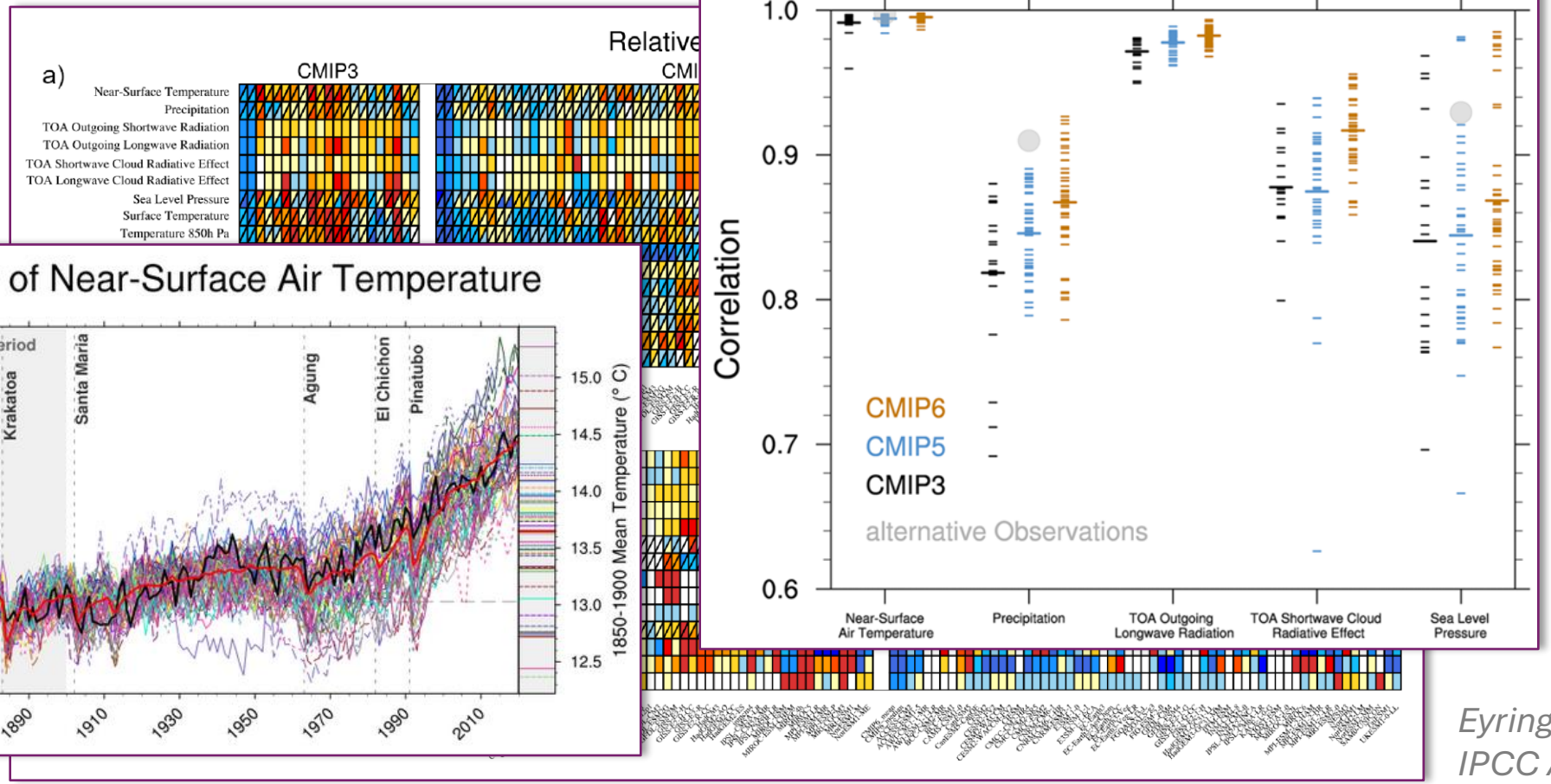
```python

from esmvaltool.diag_scripts.shar
    group_metadata,
    run_diagnostic,
    save_data,
    save_figure,
    select_metadata,
    sorted_metadata,
)
from esmvaltool.diag_scripts.shar

```

```python
def group_metadata(metadata, attribute, sort=None):
    """Group metadata describing preprocessed data by attribute.

    Parameters
    ----------
    metadata : :obj:`list` of :obj:`dict`
        A list of metadata describing preprocessed data.
    attribute : str
        The attribute name that the metadata should be grouped by.
    sort :
        See `sorted_group_metadata`.

    Returns
    -------
    :obj:`dict` of :obj:`list` of :obj:`dict`
        A dictionary containing the requested groups.
    """
    groups = {}
    for attributes in metadata:
        key = attributes.get(attribute)
        if key not in groups:
            groups[key] = []
        groups[key].append(attributes)

    if sort:
        groups = sorted_group_metadata(groups, sort)

    return groups
```

ESMValTool and IPCC

- ~ 60 scientific peer-reviewed publications
- > 20 national and international science projects, including large EU projects
- ESMValTool has been used for the production of figures for **several chapters of the Assessment Report 6 of the IPCC's WG1** (including provenance record)





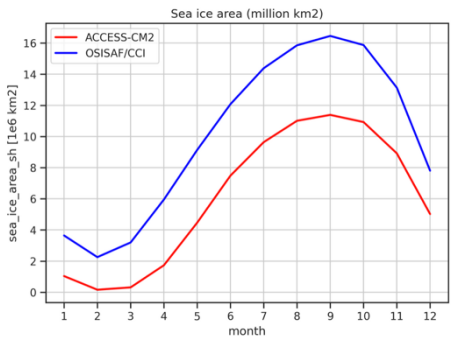
Eyring et al., Chapter 3 of IPCC AR6 WGI, 2021

ESMValTool and Climate-REF

- Rapid Evaluation Framework (REF).
- ESMValTool one of four open-source software providers.
- Diagnostics include – climate drivers of fire, climate at global warming levels, cloud properties, ENSO and climate sensitivity.
- Explore : https://dashboard.climate-ref.org/

ESMValTool

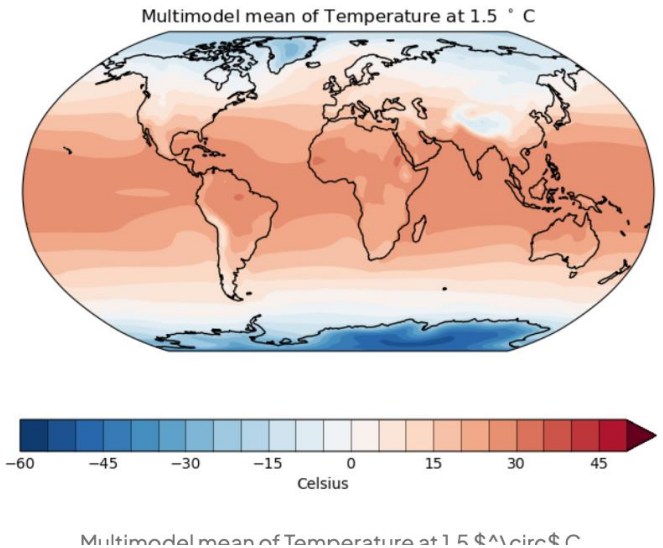
Earth System Model Evaluation Tool



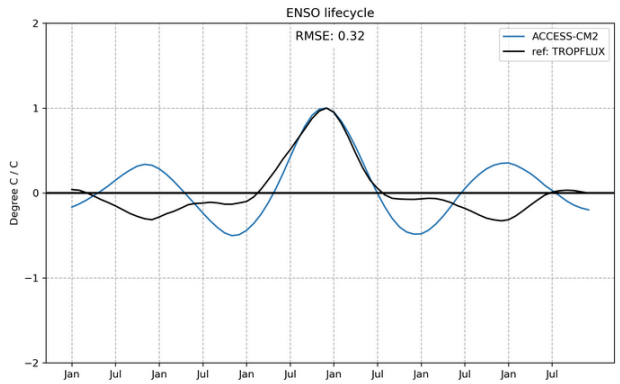
20-year average seasonal cycle of the sea ice area in million km2 from ACCESS-CM2.r10i1p1f1.gn compared with OSI-450.

Execution Group: cmip6_gn_r10i1p1f1_ACCESS-CM2

Filename: executions/recipe_20251003_125750/plots/siarea_s



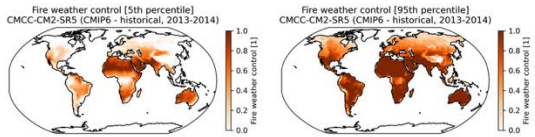
Multimodel mean of Temperature at 1.5 $^\circ$ C



Temporal evolution of sea surface temperature anomalies in the central equatorial Pacific (Niño 3.4 region average), illustrating the ENSO-associated variability.

Execution Group: cmip6_gn_r10i1p1f1_ACCESS-CM2

Filename: executions/recipe_20251002_035527/plots/diagnos



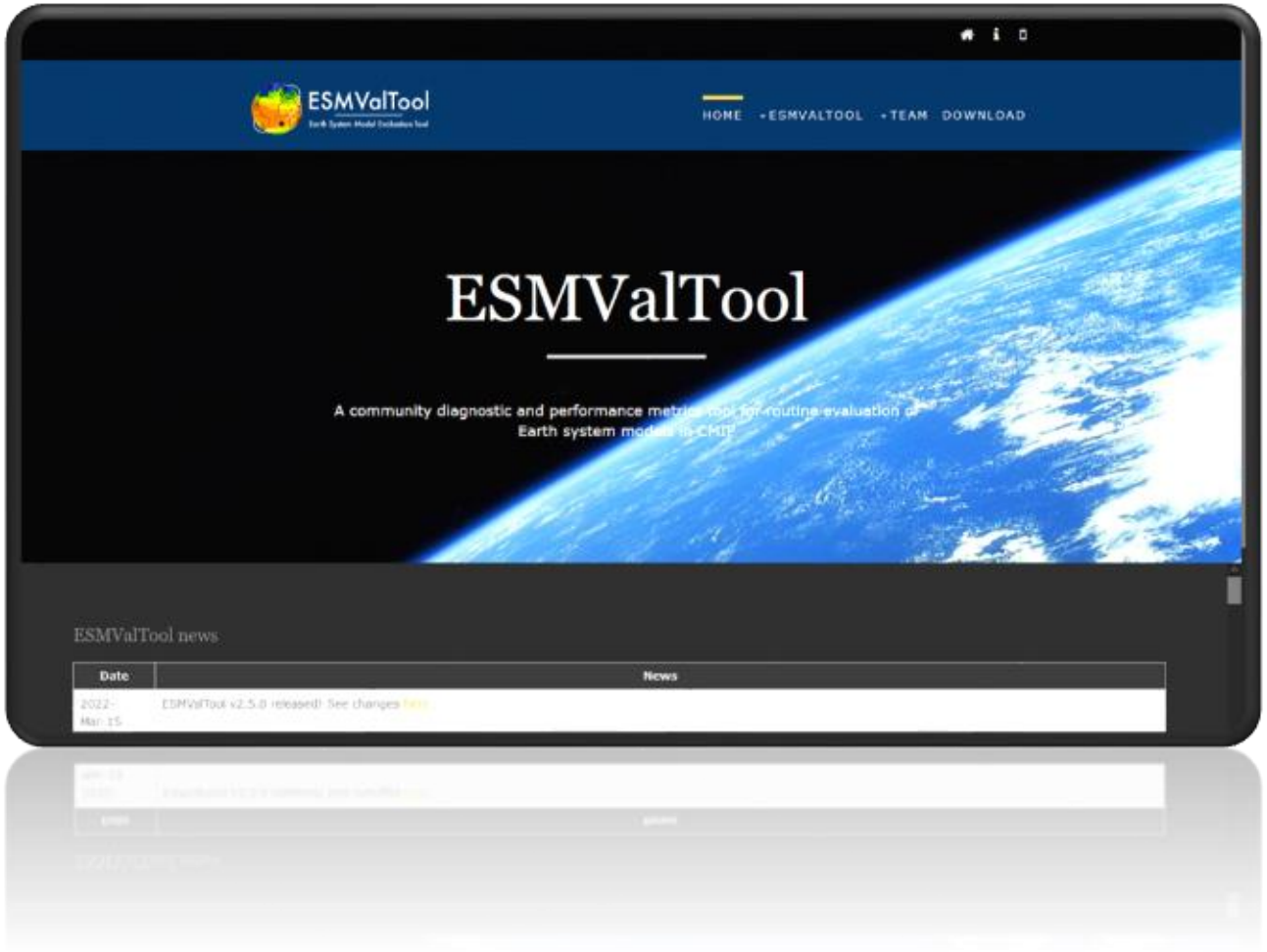
Fire weather control for the CMCC-CM2-SR5 (CMIP6-historical) for the time period 2013/2014 as computed with the ConFire model (Jones et al., 2024).

Execution Group: cmip6_gn_r10i1p2f1_CMCC-CM2-SR5

Filename: executions/recipe_20251002_035023/plots/fire_ev

ESMValTool and Climate-REF

Website



General presentation of ESMValTool:

- Important links,
- gallery,
- license,
- presentations,
- meeting notes,…

https://www.esmvaltool.org

GitHub Issues

https://github.com/ESMValGroup/ESMValTool/issues

ESMValGroup / ESMValTool Public

Notifications | Fork 135 | Star 244

Code | Issues 293 | Pull requests 98 | Discussions | Actions | Projects 1 | Security | Insights

Replace `prospector` with `ruff` integrated in Codacy
#3739 · valeriupredoi opened on Aug 16, 2024

'stable' documentation build is broken and still points to v2.7
#3490 · bouweandela opened on Dec 20, 2023 · 3

is:issue state:open | Labels | Milestones | New issue

Open 293 | Closed 1.264 | Author | Labels | Projects | Milestones | Assignees | Types | Newest

RTW configuration issues Recipe Test Workflow (RTW)
#4076 · ehogan opened 2 weeks ago · v2.13.0

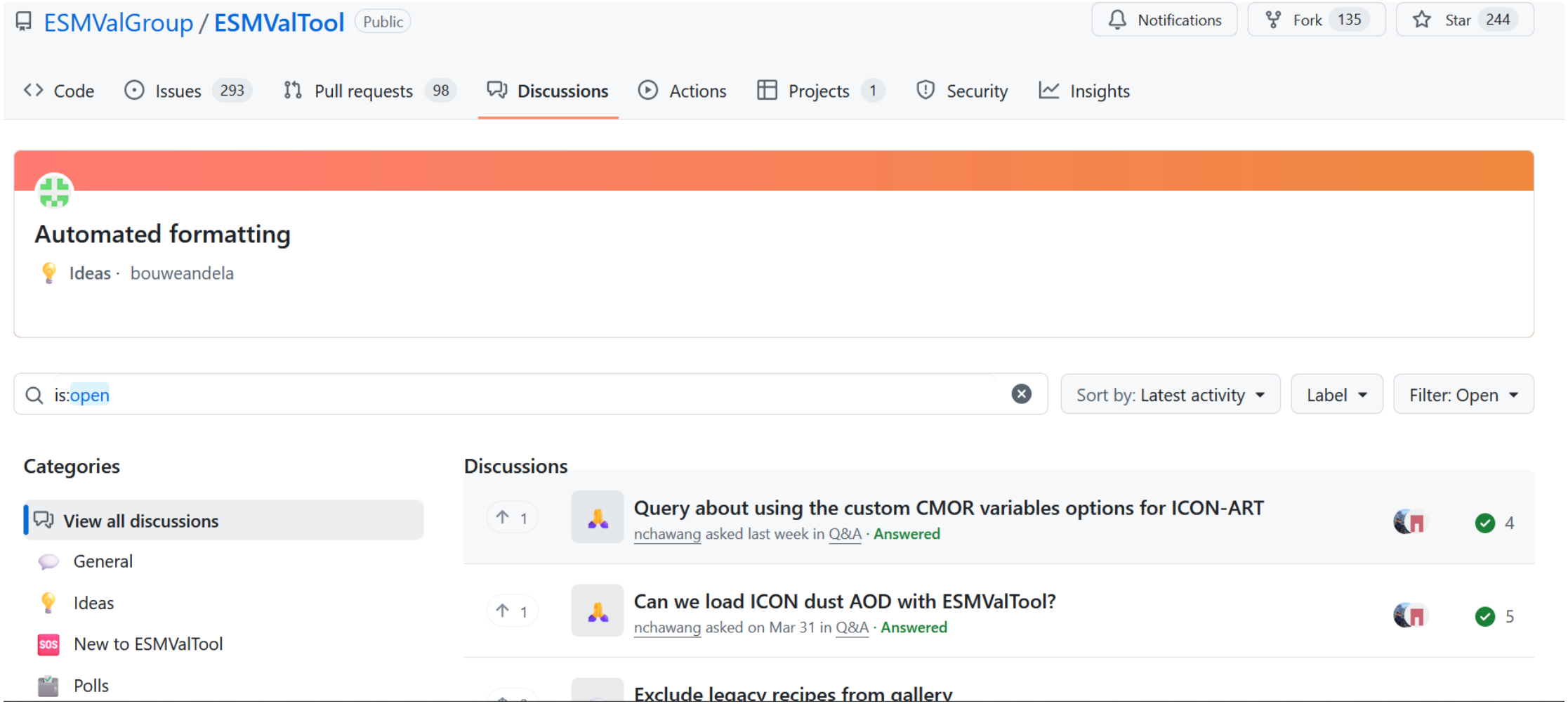
Collection of outdated Documentation parts
#4074 · bettina-gier opened 2 weeks ago

RTW config variable `BRANCH` used in both Tool and Core
#4069 · alistairsellar opened 2 weeks ago · 1 · 1

Use `ESMVALTOOL_CONFIG_DIR` rather than `--config_dir` in RTW Recipe Test Workflow (RTW)
#4053 · ehogan opened 3 weeks ago

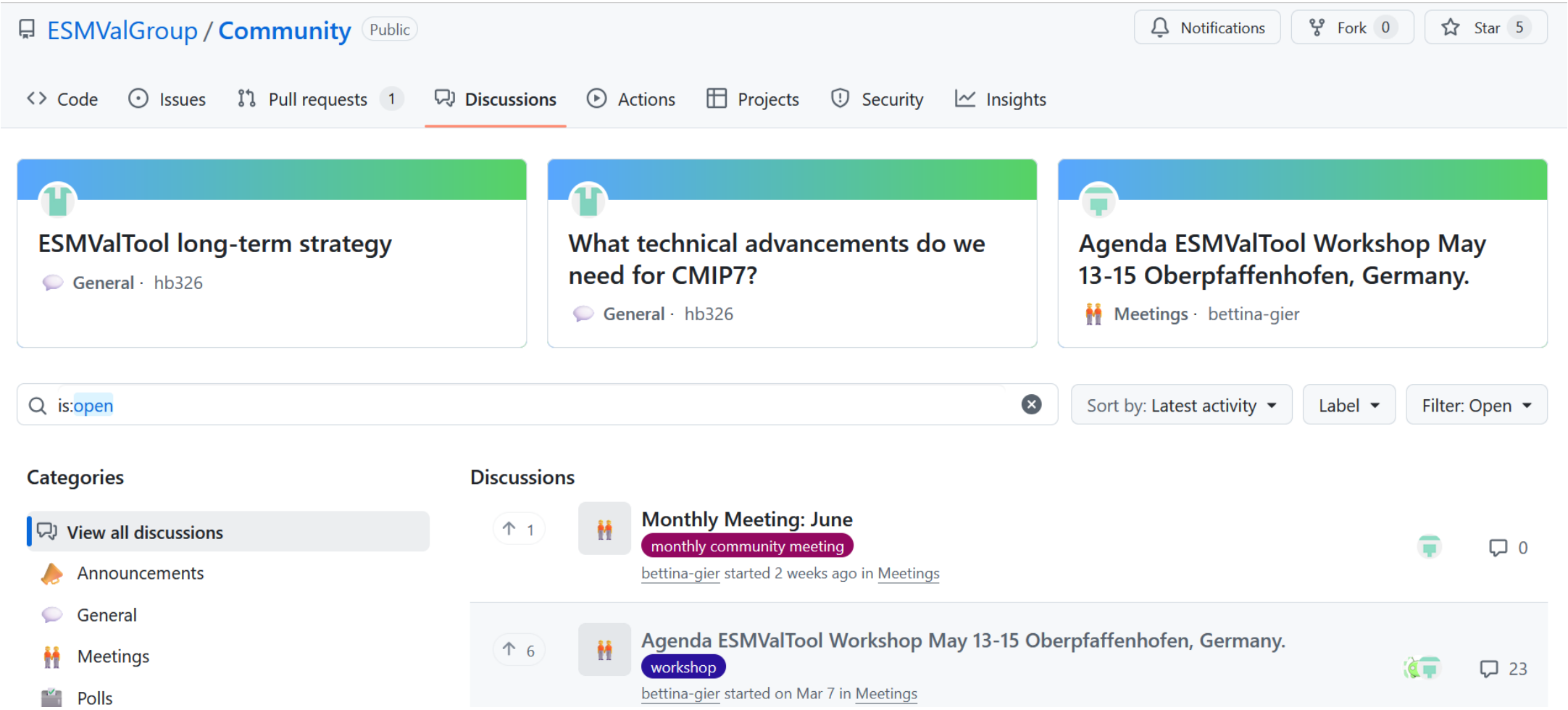
Discussions

https://github.com/ESMValGroup/ESMValTool/discussions



Community Discussions

https://github.com/ESMValGroup/Community/discussions



Documentation

https://docs.esmvaltool.org/en/latest/

ESMValTool
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Introduction | ESMValTool Functionalities | Getting started | Gallery | Recipes | More

Search Ctrl + K

Welcome to ESMValTool's documentation!

To get a first impression of what ESMValTool and ESMValCore can do for you, have a look at our blog posts Analysis-ready climate data with ESMValCore and ESMValTool: Recipes for solid climate science.

A tutorial is available on https://tutorial.esmvaltool.org.

A series of video lectures has been created by ACCESS-NRI. While these are tailored for ACCESS users, they are still very informative.

On this page

Welcome to ESMValTool's documentation!

Indices and tables

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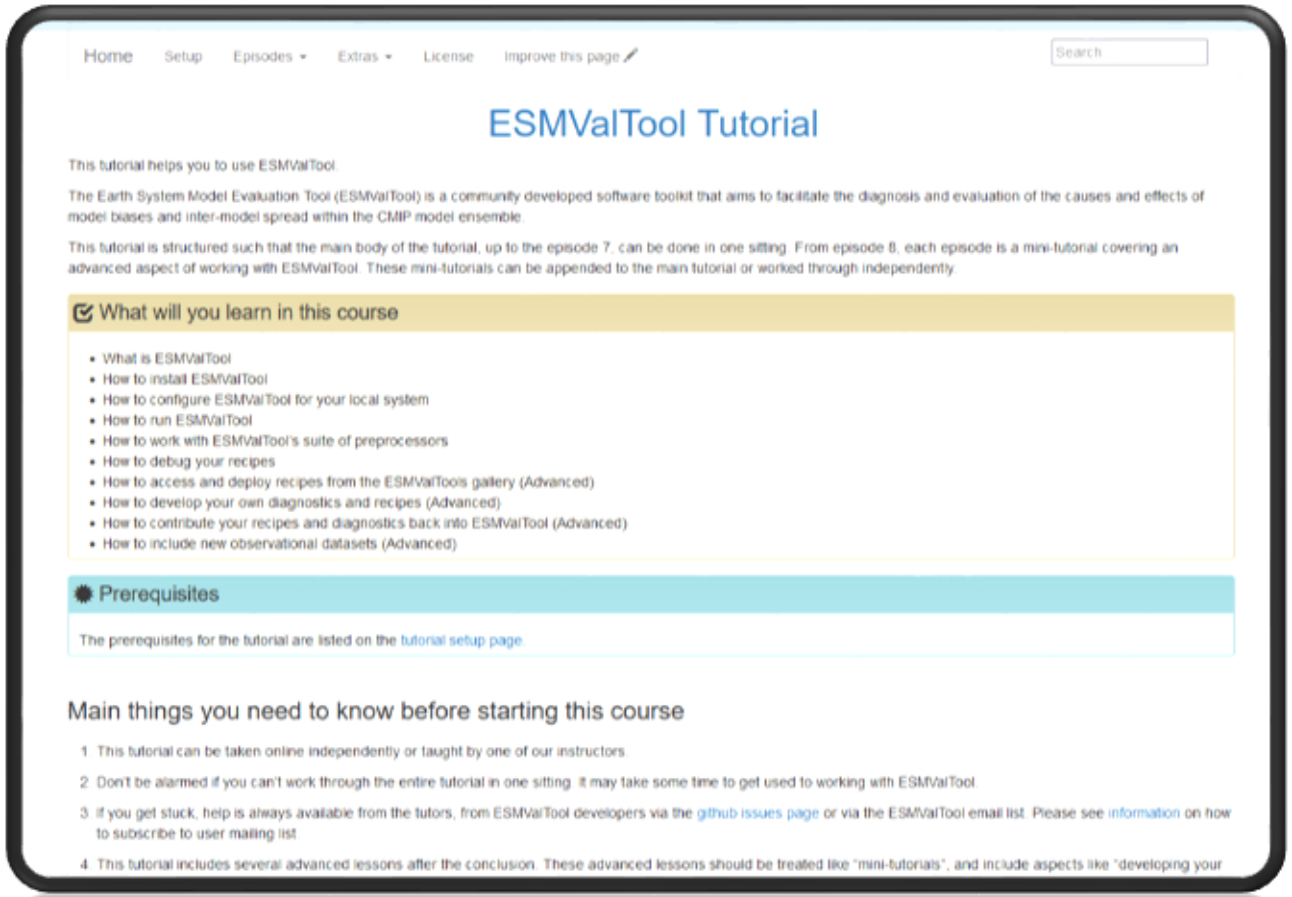
- Show Source

latest

Tutorial

https://esmvalgroup.github.io/ESMValTool_Tutorial



Home Setup Episodes Extras License Improve this page

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ESMValTool Tutorial

This tutorial helps you to use ESMValTool.

The Earth System Model Evaluation Tool (ESMValTool) is a community developed software toolkit that aims to facilitate the diagnosis and evaluation of the causes and effects of model biases and inter-model spread within the CMIP model ensemble.

This tutorial is structured such that the main body of the tutorial, up to the episode 7, can be done in one sitting. From episode 8, each episode is a mini-tutorial covering an advanced aspect of working with ESMValTool. These mini-tutorials can be appended to the main tutorial or worked through independently.

What will you learn in this course

- What is ESMValTool
- How to install ESMValTool
- How to configure ESMValTool for your local system
- How to run ESMValTool
- How to work with ESMValTool's suite of preprocessors
- How to debug your recipes
- How to access and deploy recipes from the ESMValTools gallery (Advanced)
- How to develop your own diagnostics and recipes (Advanced)
- How to contribute your recipes and diagnostics back into ESMValTool (Advanced)
- How to include new observational datasets (Advanced)

Prerequisites

The prerequisites for the tutorial are listed on the tutorial setup page.

Main things you need to know before starting this course

1. This tutorial can be taken online independently or taught by one of our instructors.
2. Don't be alarmed if you can't work through the entire tutorial in one sitting. It may take some time to get used to working with ESMValTool.
3. If you get stuck, help is always available from the tutors, from ESMValTool developers via the github issues page or via the ESMValTool email list. Please see information on how to subscribe to user mailing list.
4. This tutorial includes several advanced lessons after the conclusion. These advanced lessons should be treated like "mini-tutorials", and include aspects like "developing your

→ basic and an advanced tutorial

User Engagement Team Tasks:

- Maintaining tutorial on GitHub
- Keeping the tutorial updated with the releases
- Handling with user feedback
- Contact for tutorial hosts

Questions?

https://docs.esmvaltool.org/en/latest/introduction.html#support

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Introduction | ESMValTool Functionalities | Getting started | Gallery | Available recipes | More

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Section Navigation

Join us June 5 for Upstream: U.S. attendees are eligible for a free t-shirt. **RSVP!**

Ad by EthicalAds

Support

Support for ESMValTool can be found in ESMValTool Discussions page where users can open an issue and a member of the User Engagement Team of ESMValTool will reply as soon as possible. This is open for all general and technical questions on the ESMValTool: installation, application, development, or any other question or comment you may have.

User mailing list

Subscribe to the ESMValTool announcements mailing list esmvaltool@listserv.dfn.de to stay up to date about new releases, monthly online meetings, upcoming workshops, and trainings.

To subscribe, send an email to sympa@listserv.dfn.de with the following subject line:

- *subscribe esmvaltool*

or

- *subscribe esmvaltool YOUR_FIRSTNAME YOUR_LASTNAME*

The mailing list also has a public archive online.

On this page

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How you can contribute?

- User Engagement Team
  - Tutorial, user feedback, new user assistance.

- Tech Lead Team

- Scientific and Technical reviews
  - Indicate availability and expertise.

Thanks!

- Eso4Clima
- AI4PEX
- AIVAL
- TerraFIRMA